THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 8
2026/03/25 and 2026/03/27

1. (a) Show that $Cov(X, Y) = Cov(X, E[Y|X])$.

   (b) Suppose that for constants a and b, $E[Y|X] = a + bX$. Show that

$$b = \frac{Cov(X, Y)}{Var(X)}.$$

Solution:

(a) Note that $E[E[Y|X]] = E[Y]$ for any random variable Y. Thus, $Cov(X, E[Y|X]) = E[XE[Y|X]] - E[X]E[E[Y|X]] = E[E[XY|X]] - E[X]E[Y] = E[XY] - E[X]E[Y] = Cov(X, Y)$.

(b) From (a), $Cov(X, Y) = Cov(X, E[Y|X]) = Cov(X, a + bX) = bCov(X, X) = bVar(X)$. Therefore we have the answer.

2. Suppose X, Y and Z are discrete random variables, show that

   (a) $E(aY + bZ|X) = aE(Y|X) + bE(Z|X)$, for $a, b \in \mathbb{R}$.

   (b) If X and Y are independent, then $E(Y|X) = E(Y)$.

Solution:

(a) Take any x such that $P(X = x) \neq 0$, we have

$$\begin{aligned} E(aX + bY|X = x) &= \sum_{y,z} (ay + bz) P(Y = y, Z = z|X = x) \\ &= a \sum_{y,z} yP(Y = y, Z = z|X = x) + b \sum_{y,z} zP(Y = y, Z = z|X = x) \\ &= a \sum_{y} yP(Y = y|X = x) + b \sum_{z} zP(Z = z|X = x) \\ &= aE(Y|X = x) + bE(Z|X = x). \end{aligned}$$

(b) Take any x such that $P(X = x) \neq 0$, and by the independence of X and Y, we have

$$E(Y|X = x) = \sum_{y} yP(Y = y|X = x) = \sum_{y} yP(Y = y) = E(Y)$$

3. The joint density function of X and Y is

$$f(x,y) = \frac{e^{-\frac{x}{y}}e^{-y}}{y}, \quad 0 < x < \infty, \quad 0 < y < \infty.$$

Show that $E[X|Y = y] = y$.

Solution:

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x,y)}{f_Y(y)} \\ &= \frac{e^{-\frac{x}{y}}e^{-y}/y}{\int_0^\infty e^{-\frac{x}{y}}e^{-y}/y \; dx} \\ &= \frac{e^{-\frac{x}{y}}e^{-y}/y}{e^{-y}} \\ &= \frac{e^{-\frac{x}{y}}}{y} \\ &= \lambda e^{-\lambda x} \end{aligned}$$

if we let $\lambda = 1/y$. Therefore for a fixed y, X is just an exponential random variable with parameter $1/y$. Thus $E[X|Y = y] = (1/y)^{-1} = y$.

4. A coin having probability p of coming up heads is successively fipped until two of the most recent three flips are heads. Let N denote the number of flips. (Note that if the first two flips are heads, then $N = 2$). Find $E[N]$.

Solution:

Let X denote the number of flips such that the first head appears and consider the conditional expectation $E[N|X = x]$. The possible outcomes of the $(x+1)$-th flip and the $(x+2)$-th flip are $\{HH, HT, TH, TT\}$, respectively, where H denotes head and T denotes tail. By further conditioning on the outcomes of the $(x+1)$-th flip and the $(x+2)$-th flip, we have

$$\begin{aligned} E[N|X = x] = {} & E[N|X = x, (HH \text{ or } HT)]P(HH \text{ or } HT|X = x) \\ & + E[N|X = x, TH]P(TH|X = x) \\ & + E[N|X = x, TT]P(TT|X = x) \\ = {} & p(x+1) + (1-p)p(x+2) + (1-p)^2(x+2+E[N]) \end{aligned}$$

where the first term denotes another head immediately after the head on x-th flip, the second term denotes a tail then a head after the head on x-th flip, and the last term denote two successive tails after the head on x-th flip (i.e. the head on x-th flip doesn't contribute the end

of flipping and thus the $x+3$-th flip has no difference as the first flip). Therefore,

$$\begin{aligned}
&E[N] \\
=&\sum_{x=1}^{\infty} E[N|X=x]P(X=x) \\
=&\sum_{x=1}^{\infty}[p(x+1)+(1-p)p(x+2)+(1-p)^2(x+2+E[N])](1-p)^{x-1}p \\
=&p\sum_{x=1}^{\infty}(x+1)(1-p)^{x-1}p+(1-p)p\sum_{x=1}^{\infty}(x+2)(1-p)^{x-1}p+(1-p)^2\sum_{x=1}^{\infty}(x+2+E[N])(1-p)^{x-1}p \\
=&p\left[\frac{1}{p}+1\right]+(1-p)p\left[\frac{1}{p}+2\right]+(1-p)^2\left[\frac{1}{p}+2+E[N]\right]
\end{aligned}$$

since we know X is a geometric r.v. with $E[X]=\sum_{x=1}^{\infty}x(1-p)^{x-1}p=1/p$. So by moving $E[N]$ to one side, we have

$$E[N]=\frac{\frac{1}{p}+2-p}{1-(1-p)^2}.$$